

Notation, Abbreviations, and Hypothesis Definitions

Mathematical Symbols and Notation

Symbol	Description
N	Corpus size (total deduplicated papers)
n	Subfield paper count
T	Time span in years (for CAGR computation)
$w(a)$	Citation-weighted assertion score:
$\text{score}(H)$	$\log(1 + \text{citations}) \cdot \text{confidence}$ Aggregate evidence score for hypothesis H , range $[-1, 1]$
$S(H)$	Set of supporting assertions for hypothesis H
$C(H)$	Set of contradicting assertions for hypothesis H
$A(H)$	Set of all assertions for hypothesis H
c	Assertion confidence, range

Abbreviations and Acronyms Used

Abbreviation	Definition
AIF	Active Inference
API	Application Programming Interface
CAGR	Compound Annual Growth Rate
DCM	Dynamic Causal Modelling
DOI	Digital Object Identifier
FEP	Free Energy Principle
JSON	JavaScript Object Notation
JSONL	JSON Lines (newline-delimited JSON)
LLM	Large Language Model
NMF	Non-negative Matrix Factorization
NLP	Natural Language Processing
ORCID	Open Researcher and Contributor ID
PCA	Principal Component Analysis
POMDP	Partially Observable Markov Decision Process
RDF	Resource Description Framework
RL	Reinforcement Learning
SPARQL	SPARQL Protocol and RDF Query Language

Standard Hypothesis Definitions and Identifiers

ID	Hypothesis	Scope
H1	FEP Universality: The Free Energy Principle applies universally to all self-organizing systems	A (Core Theory)
H2	AIF Optimality: Active Inference agents achieve optimal decision-making under uncertainty	B (Tools)
H3	Markov Blanket Realism: Markov blankets correspond to real physical boundaries	A (Core Theory)

Glossary of Key Terms

Term	Definition
Active Inference	A framework in which agents minimize expected free energy to select actions, unifying perception, learning, and decision-making under the Free Energy Principle.
Assertion	A directed, confidence-scored claim linking a paper to a hypothesis (supports, contradicts, or neutral). The basic unit of evidence in the knowledge graph.
Canonical ID	The unique identifier assigned to each paper during deduplication, following the priority scheme: DOI > arXiv ID > Semantic Scholar ID >